

Nonboundary Localization Quotients of Smooth Affine $\mathfrak{sl}_2$ -Modules

The basic module $T_{f_0}M_{-1,1}$ and the family $T_{f_c}M_{a,b}$, $c < b$

September 11, 2026

Abstract

We collect the currently established structural properties of nonboundary localization quotients. These include PBW bases, exact root-mode and positive Cartan profiles, cyclicity, an explicit proper submodule at integral rank-one parameters in the deep region, annihilator preservation, and compatibility with further localization and adjoining the degree derivation. The basic example $T_{f_0}M_{-1,1}$ is treated explicitly. Simplicity at nonintegral parameters and a full submodule classification remain open here. The propositions below have direct proofs; this note is not a claim of independent peer review or formal verification.

1 Definitions, conventions, and hypotheses

All vector spaces and tensor products are over $\mathbb{C}$. We initially work without the degree derivation:

$$\mathfrak{g} = (\mathfrak{sl}_2 \otimes \mathbb{C}[t, t^{-1}]) \oplus \mathbb{C}K, \quad U = U(\mathfrak{g}).$$

Write $e_r = e \otimes t^r$, $f_r = f \otimes t^r$, and $h_r = h \otimes t^r$. Our normalization is

$$\begin{aligned} [e_r, f_s] &= h_{r+s} + r\delta_{r+s,0}K, & [h_r, h_s] &= 2r\delta_{r+s,0}K, \\ [h_r, e_s] &= 2e_{r+s}, & [h_r, f_s] &= -2f_{r+s}. \end{aligned}$$

The e 's commute with one another, the f 's commute with one another, and K is central. Here δ denotes the Kronecker delta.

Fix $a, b \in \mathbb{Z}$ with $a + b \geq 0$, and set $L = a + b + 1 \geq 1$. Let

$$S_{a,b} = \text{span}\{K, h_i \ (i \geq 0), e_i \ (i > a), f_j \ (j > b)\}.$$

Fix the character $\chi : S_{a,b} \rightarrow \mathbb{C}$ specified by

$$\chi(e_i) = \chi(f_j) = 0, \quad \chi(K) = \kappa, \quad \chi(h_i) = \lambda_i, \quad \lambda_i = 0 \quad (i > L).$$

These conditions make χ a Lie algebra character. Let $\mathbb{C}_\chi$ be the one-dimensional $S_{a,b}$ -module defined by χ , and put

$$M = M_{a,b}(\chi) = U \otimes_{U(S_{a,b})} \mathbb{C}_\chi, \quad u = 1 \otimes 1_\chi.$$

The notation κ distinguishes the central parameter from the localization index c .

Fix an integer $c < b$, and write

$$F = f_c, \quad d_* = b - c \geq 1, \quad U_F = U[F^{-1}], \quad D_F M = U_F \otimes_U M.$$

The powers of F form an Ore set, since $\text{ad } F$ is locally nilpotent on U . PBW shows that F is injective on M , so we identify M with its image in $D_F M$ and define

$$Q_c = T_F M = D_F M / M, \quad w_m = F^{-m}u + M \quad (m \geq 1).$$

Negative powers are interpreted in $D_F M$. There is no operator F^{-1} on Q_c .

A module is *smooth* if each vector is annihilated by $\mathfrak{sl}_2 \otimes t^N \mathbb{C}[t]$ for some N . An operator x is locally nilpotent if each vector is killed by a power of x ; it is locally finite if each vector spans a finite-dimensional $\mathbb{C}[x]$ -module. We call v F -torsion if $F^n v = 0$ for some n . Weights below refer to $\mathbb{C}h_0 \oplus \mathbb{C}K$, not to all h_i .

Unless explicitly stated otherwise, no assumption is made on λ_L or κ . For comparison, the original induced module satisfies [1, Theorem 1.1]

$$M_{a,b}(\chi) \text{ is simple} \iff \lambda_L \neq 0 \text{ and } \kappa \neq -2. \quad (1)$$

This criterion cannot be transferred directly to Q_c .

2 PBW structure, root modes, and weights

Proposition 2.1 (PBW model and pole filtration). Choose a PBW order with F first, followed by the families

$$f_j \ (j \leq b, j \neq c), \quad e_i \ (i \leq a), \quad h_r \ (r < 0),$$

and finally a basis of $S_{a,b}$. Let $\mathcal{X}$ denote the ordered monomials in the three middle families, including the empty monomial. Then

$$\begin{aligned} M &: \{F^q Xu : q \geq 0, X \in \mathcal{X}\}, \\ D_F M &: \{F^q Xu : q \in \mathbb{Z}, X \in \mathcal{X}\}, \\ Q_c &: \{F^{-m} Xu + M : m \geq 1, X \in \mathcal{X}\} \end{aligned}$$

are bases. In particular, Q_c is nonzero and countably infinite-dimensional. The operator F on Q_c is surjective and locally nilpotent, and

$$\ker F^r = \text{span}\{F^{-m} Xu + M : 1 \leq m \leq r, X \in \mathcal{X}\} \quad (r \geq 1).$$

Every nonzero $\mathfrak{g}$ -submodule of Q_c meets $\ker F$ nontrivially.

Proof. The first basis is the induced-module PBW theorem. It makes M free over $\mathbb{C}[F]$, hence F acts injectively. Localization gives the second basis: spanning follows by moving denominators to the left, and linear independence follows by multiplying a putative relation by a common power of F and using the first basis. Taking the quotient gives the third. The formulas for F and its kernels are now immediate. For a nonzero vector in a submodule, apply the highest power of F whose value is still nonzero. $\square$

The kernels above are vector-space filtration terms, not generally $\mathfrak{g}$ -submodules. As a module over the commuting algebra $\mathbb{C}[f_j : j \leq b]$, the quotient is a direct sum of copies of

$$(\mathbb{C}[F, F^{-1}]/\mathbb{C}[F]) \otimes \mathbb{C}[f_j : j \leq b, j \neq c],$$

indexed by the e, h PBW monomials. This is the usual first local cohomology model for a principal ideal; it does not describe the full $\mathfrak{g}$ -action by polynomial multiplication.

Lemma 2.2 (Commutation and generation by pure fractions). For $s \in U$ and $m \geq 1$, one has the finite identity

$$sF^{-m} = \sum_{j \geq 0} \binom{m+j-1}{j} F^{-m-j} (\text{ad } F)^j(s). \quad (2)$$

Moreover, $Q_c = \sum_{m \geq 1} U w_m$.

Proof. For $m = 1$, commute s past F^{-1} repeatedly; the process terminates because $\text{ad } F$ is locally nilpotent. Induction on m gives the binomial coefficient in (2).

For the generation assertion, give a numerator PBW word the degree $\rho = 2\#e + \#h$; f -factors have degree zero. Reordering does not increase this degree, and every nonzero application of $\text{ad } F$ lowers it strictly. Inducing-subalgebra factors are evaluated on u after reordering. Thus

$$Xw_m = F^{-m}Xu + M + \text{a finite sum of terms of strictly smaller numerator degree.}$$

Induct on $\rho(X)$, simultaneously for all denominator orders m . At degree zero the numerator contains only commuting f 's, so the claim is immediate. Every correction term at the next degree belongs to $\sum_{n \geq 1} U w_n$ by induction. Subtracting the corrections yields $F^{-m}Xu + M$, proving the assertion for every basis vector. $\square$

Proposition 2.3 (The negative-root profile). On Q_c , the root modes have the following properties:

index	action
$j = c$	f_j is surjective and locally nilpotent,
$j \leq b, j \neq c$	f_j is injective and not surjective,
$j > b$	f_j is locally nilpotent.

Consequently, the set of locally nilpotent negative-root modes is exactly $\{c\} \cup \{j : j > b\}$.

Proof. The first assertion is Proposition 2.1. For the second, f_j is multiplication by a free polynomial variable in the commuting-algebra model. This is injective and does not hit the constant term in that variable. For $j > b$, use $f_j u = 0$ and local nilpotence of $\text{ad } f_j$ to obtain local nilpotence on M . Since f_j commutes with F , it remains locally nilpotent on $D_F M$ and on Q_c . $\square$

Corollary 2.4 (Nonisomorphism and Hom vanishing). For $c < b$, Q_c is not isomorphic to any standard module $M_{a',b'}(\psi)$ of the form defined in Section 1. For a fixed M and distinct $c, c' \leq b$,

$$\text{Hom}_{\mathfrak{g}}(T_{f_c} M, T_{f_{c'}} M) = 0.$$

Proof. The negative-root local-nilpotence set of a standard module is the half-line $\{j : j > b'\}$, never $\{c\} \cup \{j : j > b\}$ for $c < b$. For the Hom assertion, f_c is locally nilpotent on the source and injective on the target. The image of every vector must therefore be zero. This argument also covers comparison with the boundary quotient $c' = b$. $\square$

Proposition 2.5 (Smoothness and weight structure). The module Q_c is smooth, K acts by κ , and h_0 acts semisimply with

$$\text{Supp}_{h_0} Q_c = \lambda_0 + 2\mathbb{Z}.$$

Every nonzero weight space is countably infinite-dimensional. The module has no nonzero finite-dimensional submodules or quotients, and is not integrable for the full affine algebra.

Proof. If X has r_e e -factors and r_f f -factors, then

$$h_0(F^{-m}Xu + M) = (\lambda_0 + 2m + 2r_e - 2r_f)(F^{-m}Xu + M).$$

All weights in the stated coset occur by varying $m \geq 1$ and the power of any f_j with $j \leq b, j \neq c$. Appending arbitrary powers of h_{-1} gives infinitely many linearly independent vectors of each weight. PBW also gives countability.

The induced module M is smooth: commute a sufficiently high current mode through any fixed finite PBW word until it annihilates u . Formula (2) gives

$$h_r F^{-m} = F^{-m} h_r + 2m F^{-m-1} f_{r+c}, \quad (3)$$

$$\begin{aligned} e_r F^{-m} &= F^{-m} e_r - m F^{-m-1} (h_{r+c} + r \delta_{r+c,0} K) \\ &\quad - m(m+1) F^{-m-2} f_{r+2c}. \end{aligned} \quad (4)$$

For a fixed numerator, choose $r, r + c, r + 2c$ sufficiently large; all terms vanish and the central delta is zero. The f_r 's commute with F . Hence $D_F M$, and therefore Q_c , is smooth.

Surjectivity and local nilpotence of F descend to every quotient. On a finite-dimensional vector space these properties force the space to be zero. A finite-dimensional smooth $\mathfrak{g}$ -module is trivial: there is a common high-mode annihilation bound, and the Lie ideal generated by those high modes is all of $\mathfrak{g}$, including K . But a trivial submodule of Q_c is impossible because f_{c+1} is injective. Finally, a free root mode f_j , $j \leq b$, $j \neq c$, is not locally nilpotent, excluding integrability. $\square$

Thus Q_c is a weight module, but not a finite-weight-multiplicity Harish–Chandra module. The behavior of positive Cartan modes below does not contradict semisimplicity of h_0 .

3 Positive Cartan modes and localization depth

Proposition 3.1 (Strict growth of pole order). For $1 \leq i \leq d_*$, $\zeta \in \mathbb{C}$, and $0 \neq v \in Q_c$,

$$\text{pole}_F((h_i - \zeta)v) = \text{pole}_F(v) + 1,$$

where pole order means the largest denominator exponent in the PBW expansion. In particular,

$$\ker p(h_i) = 0 \quad (0 \neq p \in \mathbb{C}[t]).$$

Thus h_i has no nonzero locally finite vectors, eigenvectors, or generalized eigenvectors.

Proof. Write the top pole term of v as $F^{-m}v_m + M$, with $v_m \neq 0$ in the span of the numerator PBW monomials containing no F . The term $F^{-m}h_i v_m$ in (3) has pole order at most m . The coefficient of pole order $m + 1$ is

$$2mf_{c+i}v_m \pmod{FM}.$$

Since $c + i \leq b$ and $c + i \neq c$, multiplication by f_{c+i} is injective on the vector-space quotient M/FM . This coefficient is therefore nonzero. Lower-pole terms and $-\zeta v$ cannot cancel it. Iteration shows that the vectors $v, h_i v, \dots, h_i^n v$ have distinct pole orders. Hence no nonzero polynomial in h_i annihilates v . $\square$

Corollary 3.2. For every standard induced module as in Section 1,

$$\text{Hom}_{\mathfrak{g}}(M_{a',b'}(\psi), Q_c) = 0.$$

More generally, Q_c has no nonzero character vector for a Lie subalgebra containing h_1 .

Proof. Such a character vector would be an h_1 -eigenvector. A homomorphism from the induced module is determined by the image of its generating character vector, which must therefore be zero. $\square$

This is not a claim that Q_c has no Whittaker vectors for *every* possible subalgebra; the condition that the subalgebra contain h_1 is essential to this argument.

Proposition 3.3 (High Cartan modes). If $i > d_*$, then $h_i w_m = \lambda_i w_m$ for every $m \geq 1$, and $h_i - \lambda_i$ is locally nilpotent on all of Q_c .

Proof. Since $c + i > b$, the correction term in (3) kills u . For a general smooth module, if $i > 0$ and $h_i w = \lambda w$, then

$$(h_i - \lambda)^N Xw = (\text{ad } h_i)^N(X)w.$$

For a fixed finite word X , each commutator with h_i either raises the index of a root factor by i , or replaces a Cartan factor by a central term, which admits no further nonzero commutator. For sufficiently large N , every nonzero term contains a root factor with arbitrarily high index. Moving it to the right can decrease its index only by sums of the fixed negative indices in the original word. It therefore still has sufficiently high index to annihilate w . This proves local nilpotence on Uw . Apply it to all w_m and use Lemma 2.2. $\square$

Consequently, the positive Cartan modes with no nonzero locally finite vectors are precisely $h_1, \dots, h_{b-c}$. This gives a second intrinsic way to read off the localization depth.

4 The basic module $Q_0 = T_{f_0}M_{-1,1}$

Here $a = -1$, $b = 1$, $c = 0$, and $L = d_* = 1$. All statements in this section allow arbitrary $\lambda_0, \lambda_1, \kappa$. The pure fractions satisfy

$$\begin{aligned} f_0 w_1 &= 0, & f_0 w_m &= w_{m-1} \quad (m > 1), \\ h_0 w_m &= (\lambda_0 + 2m)w_m, \end{aligned} \tag{5}$$

$$e_0 w_m = -m(\lambda_0 + m + 1)w_{m+1}, \tag{6}$$

$$h_1 w_m = \lambda_1 w_m + 2m f_1 w_{m+1},$$

$$e_1 w_m = -m\lambda_1 w_{m+1} - m(m+1)f_1 w_{m+2}.$$

Unlike boundary localization at f_1 , the pure-fraction recursion is controlled by λ_0 , not by λ_1 .

4.1 A canonical submodule at integral parameters

Proposition 4.1 (The e_0 -locally nilpotent part). Let

$$G = \{v \in Q_0 : e_0^N v = 0 \text{ for some } N \geq 1\}.$$

This is a $\mathfrak{g}$ -submodule, and

$$\begin{cases} 0 \neq G \neq Q_0, & \lambda_0 \in \mathbb{Z}, \\ G = 0, & \lambda_0 \notin \mathbb{Z}. \end{cases}$$

Thus Q_0 is reducible for every integral λ_0 . For nonintegral λ_0 , e_0 is injective; simplicity is not asserted.

Proof. Set $E = e_0$. If $E^N v = 0$ and $(\text{ad } E)^r(s) = 0$ for all sufficiently large r , the expansion

$$E^n(sv) = \sum_{r=0}^n \binom{n}{r} (\text{ad } E)^r(s) E^{n-r}v$$

vanishes for sufficiently large n . Since $\text{ad } E$ is locally nilpotent on U , this proves $UG \subseteq G$.

Suppose first that $p = \lambda_0 \in \mathbb{Z}$. Choose an integer $k \geq \max(1, p + 2)$, and put $x = f_0$, $y = f_1$, $\eta = \lambda_1$. The subspace $\mathbb{C}[x, y]u$ is stable under e_0, h_0, f_0 , with

$$e_0(x^j y^\ell u) = j(p - j + 1 - 2\ell)x^{j-1}y^\ell u \eta x^j y^{\ell-1}u. \tag{7}$$

Define

$$v_k = \sum_{j=0}^k a_j x^j y^{k-j} u, \quad a_0 = 1, \quad a_j = -\frac{(k-j+1)\eta}{j(p-2k+j+1)} a_{j-1}. \quad (8)$$

All denominators are nonzero, since $p-2k+j+1 \leq p-k+1 < 0$. In (7), the coefficient of $x^{j-1}y^{k-j}u$ in $e_0 v_k$ is

$$j(p-2k+j+1)a_j + (k-j+1)\eta a_{j-1} = 0.$$

Hence $e_0 v_k = 0$, and $h_0 v_k = \nu v_k$, where $\nu = p-2k \leq -2$. The rank-one commutation formula gives

$$e_0(f_0^{-m} v_k) = -m(\nu + m + 1)f_0^{-m-1} v_k.$$

The coefficient vanishes at $m = -\nu - 1$. In Q_0 ,

$$f_0^{-1} v_k + M = f_1^k w_1 \neq 0, \quad e_0^{-\nu-1}(f_1^k w_1) = 0.$$

The first equality holds because every term of (8) with $j \geq 1$ lies in M after multiplication by f_0^{-1} ; nonvanishing follows from PBW. Thus $G \neq 0$. On the other hand, take $m \geq \max(1, -p)$. Every coefficient in the successive applications of (6) to w_m is nonzero. Thus $w_m \notin G$, proving that G is proper.

Now suppose $\lambda_0 \notin \mathbb{Z}$ and $G \neq 0$. Because G is h_0 -stable and h_0 is semisimple, it contains a nonzero weight vector. Applying a suitable power of e_0 produces a nonzero weight vector $v' \in \ker e_0$. Write $h_0 v' = \nu v'$, and choose the least $N \geq 1$ with $f_0^N v' = 0$. The $\mathfrak{sl}_2$ identity implies

$$0 = e_0 f_0^N v' = N(\nu - N + 1)f_0^{N-1} v'.$$

By minimality, $\nu = N - 1 \in \mathbb{Z}$. This contradicts $\nu \in \lambda_0 + 2\mathbb{Z}$. Therefore $G = 0$. $\square$

Remark 4.2 (What this submodule does and does not prove). The submodule G is the largest submodule on which e_0 is locally nilpotent. Its restriction to $\langle e_0, h_0, f_0 \rangle \cong \mathfrak{sl}_2$ is locally finite: both root operators are locally nilpotent and h_0 acts semisimply, so the rank-one PBW theorem gives finite-dimensional cyclic submodules. The quotient Q_0/G is nonzero and e_0 acts injectively on it, since $e_0 v \in G$ implies $v \in G$. Neither simplicity of G or Q_0/G , nor length two or indecomposability of Q_0 , has been proved here.

4.2 Cyclicity, including the exceptional parameters

Proposition 4.3 (Exact cyclicity statement). The module Q_0 is cyclic for all parameters, and

$$Q_0 = U w_1 \iff \lambda_0 \notin \{-2, -3, -4, \dots\}.$$

If $\lambda_0 = -s$ with $s \geq 2$, then $Q_0 = U w_s$, while $U w_1$ is a proper submodule.

Proof. If λ_0 is not a negative integer at most -2 , every coefficient in (6), starting at $m = 1$, is nonzero. Induction on m gives $w_m \in U w_1$ for all $m \geq 1$. Lemma 2.2, whose induction covers every numerator and every denominator order, then gives $Q_0 = U w_1$.

If $\lambda_0 = -s$, the coefficient vanishes precisely at $m = s - 1$. Starting instead at w_s , repeated applications of e_0 produce all w_m with $m \geq s$, and applications of f_0 produce $w_{s-1}, \dots, w_1$. The same numerator-degree induction yields $Q_0 = U w_s$. However, $e_0^{s-1} w_1 = 0$, so $U w_1 \subseteq G$, whereas $w_s \notin G$. Proposition 4.1 proves that $U w_1 \neq Q_0$. $\square$

Two examples distinguish cyclicity from simplicity. If $\lambda_0 = -2$, then $e_0 w_1 = f_0 w_1 = h_0 w_1 = 0$, but the affine submodule $U w_1$ is infinite-dimensional: the vectors $f_1^k w_1$ are independent. If $\lambda_0 = 0$, then $Q_0 = U w_1$, yet

$$0 \neq f_1^2 w_1 \in G, \quad e_0^3 f_1^2 w_1 = 0.$$

Thus a cyclic generator need not generate a simple module.

4.3 An exact Takiff model

The subspace

$$V = \text{span}\{f_1^k w_m : m \geq 1, k \geq 0\} \cong \mathbb{C}[x, x^{-1}, y]/\mathbb{C}[x, y]$$

is stable under the nonnegative current algebra $\mathfrak{sl}_2[t]$, with all modes of degree at least two acting by zero. It is therefore a module for the Takiff algebra $\mathfrak{sl}_2[t]/t^2\mathfrak{sl}_2[t]$. Under $f_1^k w_m \leftrightarrow x^{-m}y^k$, the action is

$$\begin{aligned} f_0 &= x, & f_1 &= y, \\ h_0 &= \lambda_0 - 2x\partial_x - 2y\partial_y, \\ h_1 &= \lambda_1 - 2y\partial_x, \\ e_0 &= \lambda_0\partial_x + \lambda_1\partial_y - x\partial_x^2 - 2y\partial_x\partial_y, \\ e_1 &= \lambda_1\partial_x - y\partial_x^2. \end{aligned}$$

These formulas follow by commuting the six modes through $x^j y^\ell u$; they preserve $\mathbb{C}[x, y]$, hence descend to the quotient. They are exact formulas, not a degree truncation. The subspace V is not a submodule for the full affine algebra (for example, negative Cartan modes leave it). One cannot promote an arbitrary Takiff submodule to an affine submodule; the definition of G above is what guarantees affine stability.

5 General nonboundary indices: deep and shallow regions

5.1 Spectral-flow normalization

For $k \in \mathbb{Z}$, define the automorphism

$$\sigma_k(e_i) = e_{i+k}, \quad \sigma_k(f_i) = f_{i-k}, \quad \sigma_k(h_i) = h_i + k\delta_{i,0}K, \quad \sigma_k(K) = K.$$

Our twist convention is $x \cdot_{\text{new}} v = \sigma_k(x)v$. Tracking the inducing conditions and the localization element gives

$$(Q_c)^{\sigma_{-c}} \cong T_{f_0} M_{a+c, b-c}(\chi'), \quad \chi'(h_0) = \mu := \lambda_0 - c\kappa, \quad (9)$$

with $\chi'(K) = \kappa$ and $\chi'(h_i) = \lambda_i$ for $i > 0$. This is an isomorphism *after twisting*, not an untwisted isomorphism of the original modules. Such automorphisms also appear in [1, Section 3.1]. The useful normalized coordinates are

$$A = a + c, \quad B = b - c = d_*, \quad A + B = L - 1, \quad \mu = \lambda_0 - c\kappa.$$

Proposition 5.1 (The deep region). Suppose $c \leq -a - 1$, equivalently $d_* \geq L$. Put

$$E = e_{-c}, \quad H = h_0 - cK, \quad F = f_c.$$

These elements form an $\mathfrak{sl}_2$ -triple, $Eu = 0$, $Hu = \mu u$, and

$$Ew_m = -m(\mu + m + 1)w_{m+1}.$$

The following statements hold without assuming $\lambda_L \neq 0$:

- (i) $Q_c = Uw_1$ if and only if $\mu \notin \{-2, -3, \dots\}$. If $\mu = -s$, $s \geq 2$, then $Q_c = Uw_s \neq Uw_1$.
- (ii) The subspace

$$G_c = \{v \in Q_c : E^N v = 0 \text{ for some } N \geq 1\}$$

is a $\mathfrak{g}$ -submodule. It is nonzero and proper when $\mu \in \mathbb{Z}$, and is zero when $\mu \notin \mathbb{Z}$.

(iii) The operator E acts injectively on the nonzero quotient Q_c/G_c .

Proof. The commutator $[e_{-c}, f_c] = h_0 - cK$ gives the rank-one triple. The deep-region assumption is exactly $-c > a$, so $Eu = 0$. The formula for Ew_m follows from the rank-one commutation identity.

Here is the explicit integral-parameter construction needed to generalize Proposition 4.1; the pure-fraction recursion alone is not sufficient for all integral parameters. Set $Y_b = f_b$. On $\mathbb{C}[F, Y_b]u$, direct commutation gives

$$\begin{aligned} E(F^j Y_b^\ell u) &= j(\mu - j + 1 - 2\ell) F^{j-1} Y_b^\ell u \\ &\quad + \ell \lambda_{d_*} F^j Y_b^{\ell-1} u. \end{aligned} \quad (10)$$

Indeed, $[E, Y_b] = h_{d_*}$, the H -weight of $Y_b^\ell u$ is $\mu - 2\ell$, and the further commutators involve f_{b+d_*} , which annihilates u . Thus (10) is exactly (7) with (p, η, x, y) replaced by $(\mu, \lambda_{d_*}, F, Y_b)$.

When $\mu \in \mathbb{Z}$, choose an integer $k \geq \max(1, \mu + 2)$ and form the highest vector (8) with these replacements. It gives

$$0 \neq f_b^k w_1 \in G_c, \quad E^{2k-\mu-1}(f_b^k w_1) = 0.$$

If $d_* > L$, then $\lambda_{d_*} = 0$, and that highest vector simply equals $f_b^k u$. Pure fractions with $m \geq \max(1, -\mu)$ are not E -locally nilpotent, so G_c is proper. Stability follows from local nilpotence of $\text{ad } E$.

For $\mu \notin \mathbb{Z}$, the weight support for H is $\mu + 2\mathbb{Z}$. The highest-vector argument with the least N such that $F^N v' = 0$ in Proposition 4.1 again forces an integral highest weight, a contradiction. Thus $G_c = 0$. Injectivity on Q_c/G_c follows from $Ev \in G_c \Rightarrow v \in G_c$. Finally, the proof of Proposition 4.3 applies with μ in place of λ_0 , using Lemma 2.2 to pass from all pure fractions to the whole module. $\square$

Proposition 5.2 (The shallow region). Suppose $-a \leq c < b$, equivalently $1 \leq d_* < L$. Then $E = e_{-c}$ acts injectively on Q_c , for every value of μ . If $\lambda_L \neq 0$, the vector w_1 generates Q_c .

Proof. In this region e_{-c} belongs to the free PBW complement, so $Eu = 0$ cannot be used. Instead set $Y = e_{L-c}$. Since $L - c > a$ and $L + c > b$, one has

$$Yu = 0, \quad [Y, F] = h_L, \quad f_{L+c}u = 0.$$

There is no central term in the middle identity because $L \geq 1$. Formula (4) therefore yields

$$Yw_m = -m\lambda_L w_{m+1}.$$

For $\lambda_L \neq 0$, induction produces every w_m from w_1 ; Lemma 2.2 then proves cyclicity.

For injectivity of E , use numerator degree $\rho = 2\#e + \#h$. On the highest-degree part of a nonzero PBW expansion in Q_c , left multiplication by E adds a free e_{-c} -factor, increasing degree by two. Terms from commuting E past a numerator or a denominator have degree at most one above the original degree. In the associated graded PBW space the highest term is multiplication by a free polynomial variable, hence is injective. It cannot cancel. Therefore E is injective. $\square$

The deep-region integral obstruction therefore does not extend to all indices $c < b$. In the shallow region, whether $\lambda_L \neq 0$ and $\kappa \neq -2$ imply simplicity remains open here.

6 Extensions, annihilators, and derived functors

Proposition 6.1 (The natural extension is nonsplit). The short exact sequence

$$0 \longrightarrow M \longrightarrow D_F M \longrightarrow Q_c \longrightarrow 0$$

does not split. In fact,

$$\mathrm{Hom}_{\mathfrak{g}}(Q_c, D_F M) = \mathrm{Hom}_{\mathfrak{g}}(Q_c, M) = \mathrm{Hom}_{\mathfrak{g}}(M, Q_c) = 0.$$

If M is simple, it is the unique simple submodule of $D_F M$ and an essential submodule; consequently $D_F M$ is indecomposable.

Proof. The source Q_c is F -torsion, whereas F is injective on $D_F M$ and on M . This proves the first two Hom vanishings and excludes a section of the quotient map. The third is Corollary 3.2. For any nonzero $v \in D_F M$, a sufficiently large power of F lies in M and is nonzero. Hence every nonzero submodule of $D_F M$ meets M . If M is simple, this intersection equals M , so every nonzero submodule contains M . The remaining assertions follow. $\square$

This proves indecomposability of $D_F M$ when M is simple, not indecomposability of its quotient Q_c .

Proposition 6.2 (Annihilator preservation). In the ordinary universal enveloping algebra,

$$\mathrm{Ann}_U M = \mathrm{Ann}_U D_F M = \mathrm{Ann}_U Q_c.$$

Proof. Let $\Delta = \mathrm{ad} F$. The two-sided ideal $\mathrm{Ann}_U M$ is Δ -stable. Thus (2) shows that every element of $\mathrm{Ann}_U M$ annihilates $D_F M$. The inclusion $M \subseteq D_F M$ gives equality of the first two annihilators, and this common ideal is contained in $\mathrm{Ann}_U Q_c$.

For the reverse inclusion, let $s \in \mathrm{Ann}_U Q_c$, and choose $t \geq 0$ with $\Delta^{t+1}(s) = 0$. For every $v \in M$ and $m \geq 1$, we have $sF^{-m}v \in M$. Multiplication by F^{m+t} , using (2), gives

$$P_v(m) := \sum_{j=0}^t \binom{m+j-1}{j} F^{t-j} \Delta^j(s)v \in F^{m+t}M. \quad (11)$$

The left side is a vector-valued polynomial in m . Its finitely many coefficients have finite PBW support, so there is a common upper bound on their nonnegative F -exponents. For all sufficiently large m , their span has zero intersection with $F^{m+t}M$. Therefore $P_v(m) = 0$ for all sufficiently large integers m , and P_v is identically zero. Evaluating at $m = 0$, with binomial coefficients interpreted as polynomials, yields

$$0 = P_v(0) = F^t s v,$$

because $\binom{j-1}{j} = 0$ for $j \geq 1$. Injectivity of F on M gives $sv = 0$. Since v was arbitrary, $s \in \mathrm{Ann}_U M$. $\square$

The proposition concerns ordinary U -elements. It does not establish an analogous statement for the completed center at the critical level.

Proposition 6.3 (Functorial and derived properties). For an arbitrary U -module V , define $T_F V$ as the cokernel of the localization map, without assuming that this map is injective. Then

$$\begin{aligned} T_F V &= (U_F/U) \otimes_U V, \\ L_1 T_F(V) &\cong \{v \in V : F^n v = 0 \text{ for some } n \geq 0\}, \\ L_j T_F(V) &= 0 \quad (j \geq 2). \end{aligned}$$

The functor is right exact and is exact on short exact sequences whose three terms are F -torsion-free. In particular,

$$D_F Q_c = 0, \quad T_F Q_c = 0, \quad L_1 T_F(Q_c) \cong Q_c.$$

Proof. Ore localization is flat as a right U -module. The sequence

$$0 \longrightarrow U \longrightarrow U_F \longrightarrow U_F/U \longrightarrow 0$$

is therefore a flat resolution of the right module U_F/U . Tensoring with V identifies its degree-one homology with $\ker(V \rightarrow D_F V)$, which is precisely the F -torsion submodule. There is no higher homology. The long exact sequence gives the stated exactness on torsion-free sequences. Finally, every vector of Q_c is F -torsion. $\square$

7 Further localization and the boundary isomorphism

Proposition 7.1 (Commuting quotient localizations). Let $F = f_c$ and $G = f_j$ with distinct $c, j \leq b$. Write $D_{F,G}M$ for simultaneous localization at their powers. There are natural isomorphisms

$$T_G(T_F M) \cong \frac{D_{F,G}M}{D_F M + D_G M} \cong T_F(T_G M).$$

The analogous statement holds for any finite family of distinct free negative-root modes, independently of their order.

Proof. The modes commute, their localizations commute, and PBW makes M free in the two independent polynomial variables F, G . Inside $D_{F,G}M$, it follows that

$$D_F M \cap D_G M = M.$$

Moreover, G is injective on $T_F M$, since it still acts as a free polynomial variable. Localizing the exact sequence defining $T_F M$ at G gives

$$D_G(T_F M) \cong D_{F,G}M/D_G M.$$

Taking the quotient by the image of $T_F M$ gives the middle expression in the proposition. Its PBW basis consists exactly of terms with both exponents strictly negative. Interchanging F, G gives the other isomorphism. The same argument works for finitely many variables. $\square$

Recall the boundary result established in the earlier notes [3]: if $\lambda_L \neq 0$, then

$$T_{f_b} M_{a,b}(\chi) \cong M_{a+1,b-1}(\chi^{(1)}), \quad \chi^{(1)}(h_0) = \lambda_0 + 2, \quad (12)$$

with the other character values unchanged. The present note uses this previously proved result; it does not replace its detailed isomorphism proof.

Corollary 7.2 (Filling the interval of missing modes). Assume $\lambda_L \neq 0$. Applying one quotient localization at each of

$$f_c, f_{c+1}, \dots, f_b$$

in any order yields

$$M_{a+b-c+1,c-1}(\chi^{(b-c+1)}), \quad \chi^{(b-c+1)}(h_0) = \lambda_0 + 2(b-c+1),$$

and all other character values are unchanged.

Proof. By Proposition 7.1, arrange the operations in the order $f_b, f_{b-1}, \dots, f_c$. At each step the next mode is a boundary mode for the resulting standard module. Apply (12) repeatedly. The value of $a + b + 1 = L$ and the nonzero parameter λ_L are unchanged throughout. $\square$

For the basic example this reads

$$\begin{aligned} T_{f_1}(T_{f_0}M_{-1,1}(\chi)) &\cong T_{f_0}(T_{f_1}M_{-1,1}(\chi)) \\ &\cong M_{1,-1}(\chi^{(2)}), \quad \lambda_1 \neq 0. \end{aligned}$$

These are iterated *quotient* localizations. They are not the quotient of simultaneous localization by M alone: one must divide by the sum of the relevant partial localizations. Even if the final module is simple, simplicity of the first quotient does not follow.

8 Adjoining the derivation and the vertex-algebra viewpoint

Let

$$\tilde{\mathfrak{g}} = \mathfrak{g} \rtimes \mathbb{C}d, \quad [d, x_r] = rx_r, \quad [d, K] = 0,$$

where $x = e, f, h$. Put $\tilde{U} = U(\tilde{\mathfrak{g}})$ and $\mathcal{I}(V) = \tilde{U} \otimes_U V$. PBW gives

$$\mathcal{I}(V) \cong \mathbb{C}[z] \otimes V,$$

with actions

$$\begin{aligned} d(P(z) \otimes v) &= zP(z) \otimes v, \\ x_r(P(z) \otimes v) &= P(z - r) \otimes x_r v, \\ K(P(z) \otimes v) &= P(z) \otimes Kv. \end{aligned}$$

In particular, adjoining d is not the naive tensor-product action with unchanged current modes.

Proposition 8.1 (Compatibility with derivation induction). There is a natural $\tilde{\mathfrak{g}}$ -module isomorphism

$$\mathcal{I}(Q_c) \cong T_{f_c}\mathcal{I}(M).$$

If $0 \neq N \neq Q_c$ is a $\mathfrak{g}$ -submodule, then

$$0 \neq \mathcal{I}(N) \neq \mathcal{I}(Q_c).$$

Proof. The canonical map $\mathcal{I}(M) \rightarrow \mathcal{I}(D_F M)$ has an F -invertible target:

$$F(P(z) \otimes v) = P(z - c) \otimes Fv, \quad F^{-1}(P(z) \otimes v) = P(z + c) \otimes F^{-1}v.$$

The universal property of localization extends it uniquely to a $\tilde{\mathfrak{g}}$ -module map

$$\alpha : D_F \mathcal{I}(M) \longrightarrow \mathcal{I}(D_F M).$$

Its explicit formula is

$$\alpha(F^{-m}(P(z) \otimes v)) = P(z + mc) \otimes F^{-m}v. \quad (13)$$

The inverse sends $P(z) \otimes F^{-m}v$ to $F^{-m}(P(z - mc) \otimes v)$. These formulas respect equivalent fractions: replacing $F^{-m}v$ by $F^{-m-1}Fv$ is precisely compensated by the shift in the action of F . Thus they define inverse maps. The map α identifies the embedded copy of $\mathcal{I}(M)$ on each side.

The algebra $\tilde{U}$ is free as a right U -module, so $\mathcal{I}$ is exact. Taking quotients therefore identifies the target quotient with $\mathcal{I}(D_F M/M) = \mathcal{I}(Q_c)$, as required. Finally, $\mathcal{I}$ is faithful because its underlying vector space is $\mathbb{C}[z] \otimes V$. Applying exactness and faithfulness to $0 \rightarrow N \rightarrow Q_c \rightarrow Q_c/N \rightarrow 0$ proves the last assertion. $\square$

Thus the proper submodules G_c in the deep integral region remain proper and nonzero after adjoining d . No general preservation of simplicity in the opposite situation is claimed.

By smoothness and the fixed action of K , each Q_c is a weak module for the universal affine vertex algebra $V^\kappa(\mathfrak{sl}_2)$; see [1, Section 2]. Descending to its simple quotient requires a separate check that the defining vertex ideal annihilates the module. Smoothness alone does not imply this.

9 Open problems and limits of the present results

The following are questions, not theorems of this note.

1. **Nonintegral simplicity in the basic case.** Under $\lambda_1 \neq 0$ and $\kappa \neq -2$, the natural candidate is

$$T_{f_0}M_{-1,1}(\chi) \text{ simple} \iff \lambda_0 \notin \mathbb{Z}.$$

Only the necessary direction is proved here. The missing step is to start with an arbitrary nonzero submodule and reduce one of its vectors to a pure fraction that generates the whole quotient. Cyclicity of w_1 and injectivity of e_0 do not establish this.

2. **The corresponding deep-region criterion.** Under $\lambda_L \neq 0$, $\kappa \neq -2$, and $d_* \geq L$, does $\mu = \lambda_0 - c\kappa \notin \mathbb{Z}$ imply simplicity? A rank-one twisting or completion argument, or a direct PBW reduction, may be useful. Its applicability to these smooth modules must be proved; one cannot simply invoke a statement confined to BGG category $\mathcal{O}$.
3. **The integral-parameter submodule structure.** Determine whether G_c and Q_c/G_c are simple, whether G_c is the socle, whether their extension splits, and whether Q_c has finite Jordan–Hölder length. At present, neither length two nor indecomposability is asserted.
4. **Simplicity in the shallow region.** Here e_{-c} is injective for all μ , so the deep-region integral obstruction disappears. Does simplicity of the original module suffice, or are there other resonance conditions?
5. **Parameter isomorphisms and degenerations.** The level κ , support $\lambda_0 + 2\mathbb{Z}$, negative-root local-nilpotence set, and positive-Cartan local-finiteness profile give necessary invariants. For $i > d_*$, the generalized eigenvalue λ_i is an additional invariant. A complete parameter classification, especially for $\lambda_L = 0$ or $\kappa = -2$, remains unresolved.
6. **Critical-level central reduction.** At $\kappa = -2$, study the completed center, its central reductions, their compatibility with localization, and the resulting simple quotients. Proposition 6.2 does not settle completed-center questions.
7. **Multiple missing modes and twisted localization.** Proposition 7.1 suggests a family indexed by finite sets of free modes. One may also study generalized conjugation

$$\Theta_\xi(s) = \sum_{j \geq 0} \binom{\xi}{j} (\text{ad } F)^j(s) F^{-j}, \quad \xi \in \mathbb{C},$$

on $D_F M$, where the sum is finite for each $s \in U$. This construction requires F to be invertible; it is not directly an operation on Q_c .

10 Sources and verification record

The original induced modules, their simplicity criterion, spectral-flow automorphisms, and the smooth-module vertex-algebra correspondence are drawn from [1]. The localization results of [2] provide relevant comparisons: Theorem 4.3 exhibits an integral rank-one reducibility condition $J - iK \in \mathbb{Z}$ for imaginary Verma/free-field modules, and Section 5 constructs primitive vectors and proves the corresponding nonintegral simplicity statements. Those modules differ from the present ones, so their sufficiency theorems are not used as proofs of our open simplicity questions.

The boundary isomorphism and the derivation-induction framework were established in the earlier project notes [3]. This document is an English organization of the current nonboundary results, not a revision of those earlier boundary proofs.

The accompanying computation work/`check_nonboundary_takiff.py` recorded 5,866 exact rational checks of the six Takiff generators' bracket relations, the highest-vector recurrence at integral parameters, explicit locally nilpotent vectors, and pure-fraction resonance. The checks include $\lambda_1 = 0$ and do not truncate polynomial degree in the action formulas. They nevertheless test only finitely many parameters and vectors and do not replace the general proofs. No new Danus run was used for these nonboundary results. No independent audit or Lean/Coq formal verification is claimed.

References

- [1] V. Futorny, X. Guo, Y. Xue, and K. Zhao, *Smooth representations of affine Kac–Moody algebras*, arXiv:2404.03855v2 (2024; journal version 2025). [arXiv full text](#).
- [2] V. Futorny, D. Grantcharov, and R. A. Martins, *Localization of free field realizations of affine Lie algebras*, arXiv:1404.7148v2 (2015). [arXiv full text](#).
- [3] Earlier project working notes, *Boundary localization: advisor note* (September 2026) and *Boundary localization and derivation extension* (September 8, 2026). English source files: `boundary_localization_advisor_note_en.tex` and `boundary_derivation_extension_en.tex`.